

Weekly Report (Work Plan): Acceptance Criteria for the Uncertainty Propagation

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Question

Which acceptance criteria connect the propagated kernel-extraction error, for the ladder step positions and the 5% tolerance band, to the experimental design?

Hypothesis

Acceptance should be defined not by the absolute wavelength endpoints but by three quantities — the spread of the step boundaries in *logarithmic power ratio*, the number of grid points required at 1% precision, and the probability of missing a step (the previous draft compared wavelength endpoints with the dimensionless grid ratio 1.227, which is a comparison of unlike quantities and was wrong).

Evidence / Interpretation

- The inputs are already fixed by the figure cross-check (E3): the sideband reproduces to within 1.0% with $r = 1.00$ and matching peaks, while the ZPL side carries a $\sim 2\%$ read-off error on the Debye–Waller factor from the axis calibration. The ZPL peak height is unusable because of clipping.
- Figure 1 shows why the criterion must be stated in log power. The quantity at risk is the position of the six vertical boundaries, and they are not equally exposed: the pair at 1.5145 and 1.5191 is separated by 0.3% in power, so the $N = 6$ plateau between them is already unreachable before any error is added, whereas the boundaries at 1.8646 and 4.2989 stand clear by factors of 1.23 and 1.19. Whether a propagated error matters is therefore a question about spacing in $\log I$, not about nanometres.

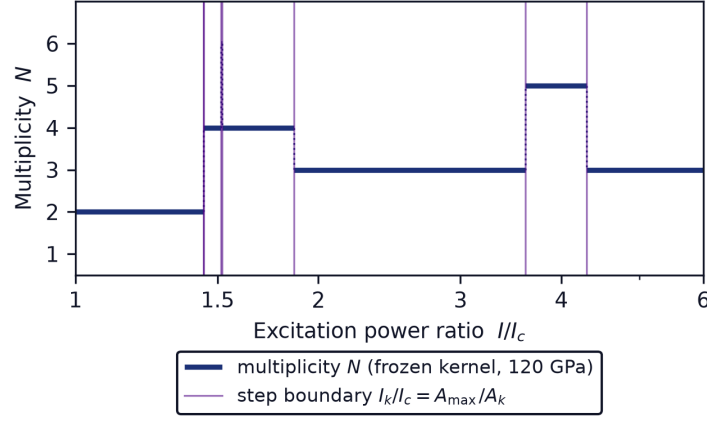


Fig. 1 Multiplicity ladder $N(I/I_c)$ computed from the frozen v3 kernel at 120 GPa. Vertical lines mark the step boundaries $I_k/I_c = A_{\max}/A_k$ at 1.4414, 1.5145, 1.5191, 1.8646, 3.6079 and 4.2989. The $N = 6$ plateau spans only 0.3% in power and is not resolvable.

Gap / Failure

- Because the Monte Carlo has not been run, the question of whether these boundaries are stable under propagated error remains unanswered in this report; the figure shows their nominal positions only.
- The knobs of the v1 `default_randomiser` are unusable, as the quantities they addressed no longer exist. The randomiser must be replaced wholesale before the propagation can start.

Next Step

Implement the Monte Carlo with 1% sideband and 2% DWF inputs, and report its output as the three quantities: step-boundary spread in log power ratio, required number of grid points, and step-miss probability.

References

1. K. O. Ho, C. Dailledouze, V. Žalandauskas *et al.*, “Optical Stability and Photophysics of NV Centers in Diamond up to 120 GPa,” arXiv:2606.02399v1 [quant-ph] (2026). Source of the optical kernel $A(\lambda, P)$, reconstructed from Fig. 1(e).
2. A. Dréau *et al.*, “Avoiding power broadening in optically detected magnetic resonance of single NV defects for enhanced dc magnetic field sensitivity,” *Phys. Rev. B* **84**, 195204 (2011). Source of the branching exponent $E = 3$ and $\rho^* = 1/5$.